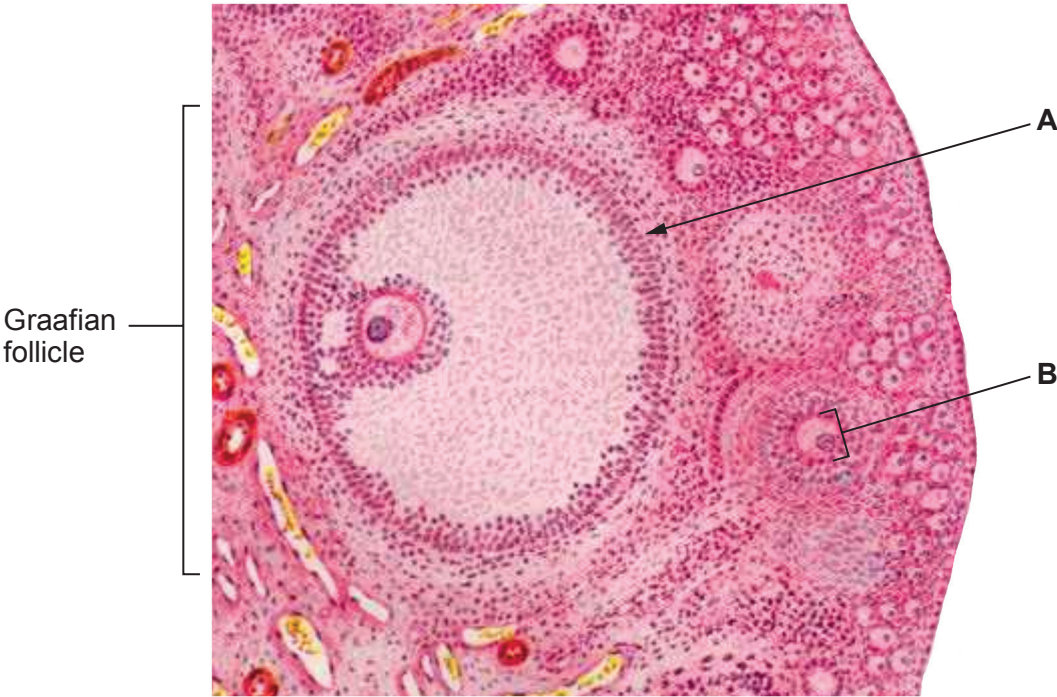


Answer **all** questions.

1. The photomicrograph below shows a section through part of a human ovary.



(a) (i) Name the structures **A** and **B**. [2]

A

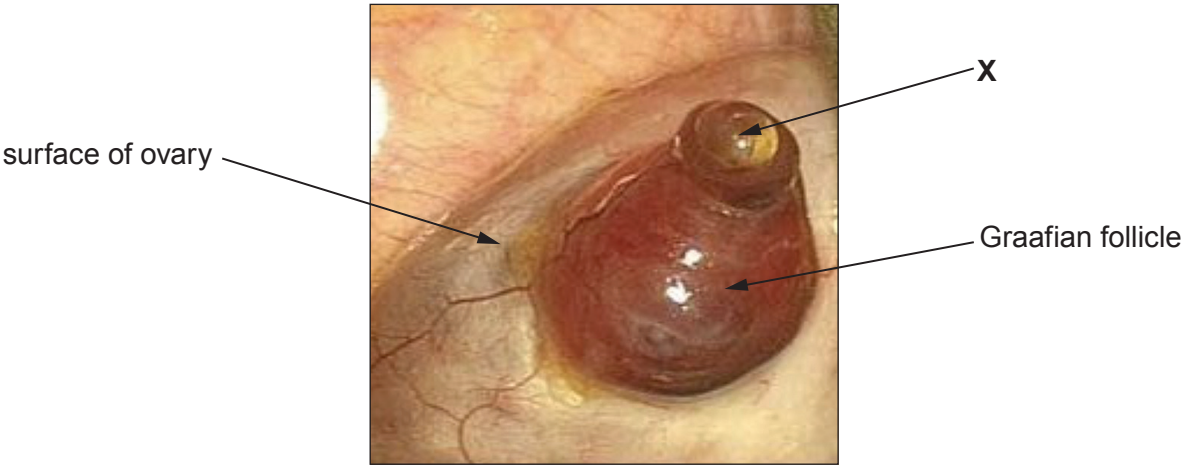
B

(ii) The volume of a primary follicle is 0.00005mm^3 and the volume of a mature Graafian follicle is $4.19 \times 10^3\text{mm}^3$. By how many times is the volume of the Graafian follicle bigger than the volume of the primary follicle? Express your answer in standard form. [3]

Graafian follicle is times bigger than the primary follicle



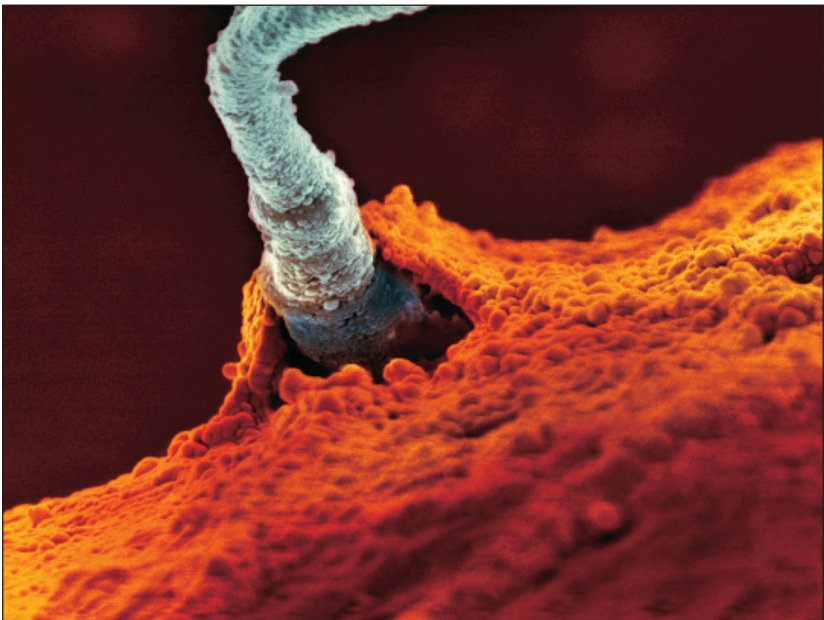
In the human female, the first half of the meiotic division takes place just before ovulation. The photograph below shows the surface of an ovary at ovulation.



(b) Structure **X** contains the secondary oocyte. Name the **two** outer layers of structure **X**. [1]

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The scanning electron micrograph shows a sperm penetrating the surface of structure **X**.



(c) Describe how the sperm is able to penetrate the outer layers of this structure. [2]

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(d) In vitro fertilisation (IVF) is a technique available to help people with fertility problems. During IVF, secondary oocytes are removed from a woman's ovaries and are fertilised with sperm in a laboratory.

- (i) There is a risk that during fertilisation more than one sperm enters the secondary oocyte. The risk is increased if the secondary oocyte is not fully mature. There is a 6% increased chance of more than one sperm entering the secondary oocyte with IVF.

State how the secondary oocyte usually prevents the entry of more than one sperm and suggest a reason why there is an increased risk with IVF. [2]

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- (ii) The developing embryo is not transferred into the uterus until three days after IVF. Using your knowledge of fertilisation and implantation explain why this delay is needed. [3]

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